Anticipating Network Contagions: When Directedness Outperforms Connectedness

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Abstract

When a shock propagates through a network, responders can reactively reinforce the worst-hit unit or anticipatorily support the quiet transmitter exporting the cascade. We investigate when anticipation succeeds, delivering a transparent, falsifiable diagnostic framework. Calibrating non-negative vector autoregressive surrogates across seven empirical contagion networks (epidemic, financial, environmental, transport, conflict) and diagnosing structures via generalized forecast error variance decompositions, we reveal a fundamental directed-asymmetry regularity: transmitter-targeting outperforms reactive control if, and to the degree that, a system is directionally asymmetric. Systemic controllability is dictated by this directed asymmetry rather than overall network connectedness magnitude. We isolate this property using a per-case symmetrization null and evaluate it via a prediction-first protocol. Across five pre-committed out-of-domain tests, predictions were confirmed twice, refined once, and falsified twice (yielding three total confirmations including a held-out COVID-19 network). The diagnostic successfully recovers the historical origin in five of seven networks, a significant recovery rate against a random null ($P \approx 10^{-4}$) that remains robust where transmitters and loudest nodes diverge ($P \approx 1.5 \times 10^{-3}$). Ultimately, this provides an efficient, reproducible diagnostic mapping where anticipatory intervention has the structural leverage to succeed.

Introduction

In many systems of practical concern, such as financial networks, epidemics, and environmental or infrastructure systems, the outcome at one unit depends fundamentally on the treatment of others. This interference voids the stable unit treatment value assumption (SUTVA) that underpins classical causal inference [1, 2, 3, 4], fundamentally altering the decision problem. A controller operating under a limited budget must choose *where* to intervene, deciding whether to *react* to realized damage or to *anticipate* its spread. Reaction forms the default logic of most triage protocols, allocating resources to the most severely damaged unit (the *loudest-node heuristic*). Anticipation, conversely, requires modeling the network to support the units propagating the cascade, which are often quiet nodes experiencing minimal localised damage.

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While targeting structurally important nodes possesses a rich history in network control and epidemic immunisation [12, 13, 15, 16, 17], our inquiry is narrower and strictly operational: we ask when anticipation outperforms reaction, and what structural property dictates that advantage. Closest to this inquiry, structure-aware optimal control has recently been shown to outperform static centrality heuristics for allocating time-varying interventions [21]. However, that result operates within an *undirected* regime lacking a formal transmitter-versus-receiver distinction. Our work advances a complementary, strictly directed claim: the operative axis of control is directed asymmetry, a property we isolate on a per-case basis via a symmetrization null. To operationalise this, we adopt the econometric connectedness apparatus [22]. This primitive has been developed extensively to measure systemic risk and directed information flow, featuring extensions into Bayesian source/sink ranking [24], wide elastic-net volatility-spillover networks [25], and broader directed-information foundations [26]. However, it has historically functioned strictly as a descriptive tool for estimation, never as an active test of network controllability. That interventional control layer constitutes our primary methodological contribution.

We demonstrate that the value of anticipatory control is governed by a single structural property of the contagion network. We define a *transmitter* as a unit that, within a fitted dynamical model, exports more forecast-error variance than it imports. Conversely, the *loudest unit* is defined as the node experiencing the largest realized damage, representing the maximal current value of the modeled stress variable rather than an independent impact metric. Throughout this work, "directed" denotes predictive directional influence in a Granger-causal or forecast-error-variance sense, remaining distinct from an interventional causal effect (see Methods).

Our central claim is a falsifiable **directed-asymmetry regularity**: a transmitter-targeting controller outperforms reactive, damage-weighted control precisely when the network is directionally asymmetric (i.e., when the loudest unit and the transmitter differ). This advantage collapses to the baseline when the network is symmetric, weakly coupled, or when the transmitter is itself the loudest unit. Crucially, the deciding parameter is the *directedness* of the spread rather than the overall magnitude of its connectedness. Two networks can exhibit identical aggregate connectedness yet differ entirely in controllability, a separation we validate empirically using a near-matched connectedness pair alongside a formal symmetrization null. An independent finding within power-systems engineering reinforces this principle: the influence graph governing how cascading outages physically propagate is demonstrably distinct from the underlying physical grid topology [23]. This represents a different-domain instance of the same dissociation between operative directed influence and raw connectivity that our diagnostic formalises.

To test this regularity, we apply a single, transparent analytical pipeline that sequentially executes surrogate calibration, structural diagnosis, and simulated interdiction uniformly across all evaluated systems. We frame the result strictly as a falsifiable regularity and a diagnostic tool rather than a universal theoretical framework. Consequently, we commit to structural predictions prior to computing each network and explicitly report our empirical misses. The underlying components of this pipeline are individually standard: a non-negative vector autoregression (VAR), Diebold–Yılmaz generalized forecast error variance decomposition (GFEVD) [5, 6], and a cross-entropy model predictive controller (MPC). However, our novelty lies in the two distinct verification objects these components enable. The first is a per-case symmetrization null that mathematically isolates the contribution of directed asymmetry from that of raw connectedness magnitude. The second is a pre-committed confirm–refine–falsify protocol that equips the regularity with a transparent out-of-sample record and a rigorously mapped empirical scope. Finally, this work was originally motivated by a geometric interpretation of network dynamics, a conceptual perspective we term Last Straw Analytics, wherein interference is viewed as the leakage of a unit's Markov blanket and a cascade

as the loss of a separating boundary. We summarize this viewpoint in the Discussion strictly as an interpretive lens rather than a validated empirical claim.

The directed-asymmetry regularity and its isolation

Pipeline. For each system, we fit a non-negative first-order VAR to the multivariate stress series (whether incidence, depreciation, pollution, delay, or fatalities), establishing a *calibrated linear surrogate* rather than a mechanistic model. This linear abstraction provides a domain-agnostic, computationally tractable substrate for control, allowing us to apply an identical analytical pipeline uniformly across highly heterogeneous systems. From the fitted coefficient matrix Φ and residual covariance, we compute the generalized forecast error variance decomposition and the Diebold–Yılmaz connectedness table [5, 6]. This calculation yields each unit's *to*, *from*, and *net* directional connectedness, where the net-positive unit displaying the largest net connectedness is classified as the transmitter (Figure 1).

We then pose an interdiction problem on the surrogate: utilizing a fixed per-step budget, the controller allocates support to attenuate a cascade. We compare a reactive, damage-weighted loudest-node heuristic against static transmitter-targeting and a cross-entropy model predictive controller (MPC) [9]. Because fitted surrogates vary in their native stability (some are inherently stable, while others, such as the wildfire-smoke fit, are already supercritical at $\rho \approx 1.28$), we rescale each to a common, mildly supercritical regime ($\rho = 1.05$–1.06). This ensures that the controllers face a comparable cascade environment. Consequently, every reported effect size remains a property of this calibrated supercritical surrogate rather than a measured real-world intervention. We demonstrate below that the qualitative regularity, rather than the absolute interdiction magnitude, represents the mathematically robust object.

Isolating the asymmetry. To rigorously test whether the transmitter-targeting advantage is driven by directed asymmetry rather than raw connectedness, volatility, or sheer coupling magnitude, we introduce an in-surrogate symmetrization null. We interpolate the fitted Φ matrix toward its symmetric projection, $\frac{1}{2}(\Phi + \Phi^\top)$, and dynamically recompute the controller performance. If the directed topology serves as the active structural mechanism, the transmitter-targeting advantage must decay as the network approaches symmetry. Across all confirmatory cases, this advantage collapses exactly as expected. Furthermore, as a strict diagnostic check, this performance edge remains entirely absent in the baseline systems where the regularity predicts no operational control advantage.

Connectedness does not predict controllability. The clearest illustration of this dissociation emerges from a matched pair of networks displaying nearly identical total connectedness. The 2023 wildfire-smoke network (82%) is inherently directionally asymmetric, characterized by an upwind transmitter and a downwind, loudest-node receiver (Figure 2). Within this system, transmitter-targeting reduces the surrogate cascade approximately 2.5× more effectively than the loudest-node heuristic (+10% versus +4%). This advantage elegantly collapses under the symmetrization null. However, because its block-bootstrap confidence interval includes zero, we treat this specific network pair as structurally illustrative, deferring formal statistical weight to the equity and COVID-19 networks detailed below.

In sharp contrast, the 2013–2014 flight-delay network is almost identically connected (77%) yet remains structurally near-symmetric. Here, every evaluated controller performs near the no-

action baseline ($\leq 8\%$) because the system's native state begins below the directedness threshold on the symmetrization null, leaving no clean directed edge to exploit. Ultimately, these two cases demonstrate that equivalent connectedness can yield entirely opposite systemic controllability. This fundamental divergence is decided exclusively by directedness, which the per-case symmetrization null isolates and measures directly.

The clearest case: directed control where reaction is counterproductive. The 2008 global equity crash (aggregating eight national indices weekly from 2007–2010) represents the most highly connected financial network we examine (81%), and an unambiguously directed one. The United States, representing the documented subprime and Lehman epicenter, operates as the dominant net-transmitter (+21% net), while indices in Hong Kong and Japan function as receivers. Here, the contrast between anticipation and reaction becomes highly pronounced. Transmitter-targeting reduces the surrogate cascade by approximately 42%, whereas the reactive loudest-node heuristic proves actively *counterproductive* (-7%). By allocating resources to the loudest heuristic, the controller merely reinforces downstream sinks while the upstream source continues transmitting unhindered.

This entire advantage reverses under the symmetrization null ($+49 \rightarrow -23$ as the kernel is symmetrized), providing the definitive signature of a directed mechanism. Together with the held-out COVID-19 network, the equity crash represents one of only two networks whose block-bootstrap advantage intervals explicitly exclude zero (Supplementary Information). Crucially, both are systems featuring a quiet, documented origin that remains structurally distinct from the loudest casualty. We examine the equity crash as a robust, directed financial confirmation independent of the six out-of-domain predictive tests detailed below, serving as the seventh network in our ground-truth recovery audit.

Benchmark. We evaluate this diagnostic approach against competitive, high-performing alternatives rather than trivial baselines. On a held-out, empirical COVID-19 state network, structurally directed influence measures reduce the surrogate cascade by 35–65%. These measures include Diebold–Yılmaz net connectedness, impulse-response spillover [8], and VAR out-strength. By comparison, undirected correlation centrality achieves only a 5% reduction, while reactive triage yields a negligible 0.2% attenuation. Notably, a computationally intensive cross-entropy MPC reduces the cascade by 30% (Figure 3), demonstrating that our static directed diagnosis mathematically matches or exceeds the dynamic planner on this specific empirical network. Furthermore, these directed measures recover the true support targets with rank correlations ranging from 0.5 to 0.8, compared to -0.1 for the undirected baseline. These results underscore that *directedness* itself, rather than the technical nuances of a specific estimator, drives the structural control advantage. To confirm that this advantage is not an artefact of a within-paradigm comparison, we additionally evaluated three cross-paradigm controllers drawn from the source-identification and network-control literatures—effective-distance source centrality [14], collective influence [13], and structural-controllability driver nodes [12]—as competing interventions on the two anchor networks (Supplementary Information). Transmitter-targeting is competitive with the strongest of these: it matches collective influence on the 2008-equity network and trails it modestly on the held-out COVID-19 network, while every directed or source-structure method decisively outperforms reactive, damage-weighted control. Consistent with our central claim, the operative property is directedness itself; no method that ignores source structure is competitive.

Why directedness governs control. The mechanism underlying this regularity is structurally intuitive. Because a transmitter sits upstream within the directed flow of a cascade, attenuating its initial stress removes variance from all subsequent downstream units. Conversely, the loudest unit typically operates as a structural *sink*, accumulating damage it did not natively originate. Intervening at such a sink merely treats localised symptoms without staunching the cascading systemic inflow. When a network is symmetric or weakly coupled, no consistent upstream axis exists to exploit. Consequently, no single allocation strategy can systematically dominate, and anticipation fails to outperform reactive triage. Finally, if the systemic origin of the cascade is simultaneously the worst-hit unit, the transmitter and the loudest node coincide, inherently collapsing the anticipatory advantage back to the reactive baseline.

Six out-of-domain tests

A structural pattern qualifies as a scientific regularity only if it withstands risky out-of-sample predictions. Consequently, we subject our central thesis to six rigorous out-of-domain tests: one held-out dataset and five prediction-first evaluations spanning four distinct empirical domains. For the five prediction-first tests, we committed our structural hypotheses to a written record prior to computing the networks. While not formally pre-registered with an external public timestamp, these predictions were strictly fixed before any data analysis commenced. Collectively, these six tests span the three operational outcomes admitted by the regularity, yielding three confirmations, one refinement, and two falsifications (Figure 4). Specifically, the initial held-out COVID-19 network (reutilized below for our validity analyses) robustly confirms the regularity, while the five prediction-first tests split into two confirmations, one refinement, and two falsifications.

Confirmations. For the 1997 Asian financial crisis (aggregating nine currencies), we predicted prior to network construction that the quiet historical origin (the Thai baht) would act as the systemic transmitter, while the loudest casualty (the Korean won) would act as the structural receiver. The empirical network unambiguously confirmed this hypothesis: transmitter-targeting outperformed the loudest-node heuristic by a factor of approximately 2.3 (+13% versus +6% on the surrogate cascade). We observed a mathematically parallel dynamic in the 2023 wildfire-smoke network (seventeen U.S. states). Here, an upwind state (Wisconsin) acted as the transmitter while the worst-hit state (Michigan) served as the receiver, yielding a comparable 2.5× performance advantage for the anticipatory transmitter-targeting controller.

Refinement. For U.S. influenza propagation across fourteen states (2010–2019), the network exhibited strong directedness, but the empirically identified net-transmitter (Texas) simultaneously functioned as the loudest state. Because the structural gap between the loudest unit and the transmitter was entirely absent, transmitter-targeting outperformed the loudest-node heuristic only marginally ($\approx 1.1\times$). This outcome formally refines the regularity, demonstrating that the operative driver of the control advantage is the explicit divergence between the loudest unit and the transmitter, rather than directed asymmetry in isolation. Coincident transmitters (as observed in influenza) and near-symmetric systems (as observed in flight delays) thus represent two distinct structural pathways through which the anticipatory edge vanishes.

Falsifications and empirical boundaries. Two of our committed predictions failed to match our initial hypotheses. We report them here transparently to delineate the rigorous boundary

conditions of our regularity. First, for U.S. flight-delay propagation during the 2013–2014 polar-vortex winter (eighteen airports), we predicted a directed front seeded by a weather-exposed hub. Instead, the empirical network proved highly connected (77%) but structurally near-symmetric: the net-transmitter emerged as a warm Sun Belt airport, while the heavily disrupted weather hubs acted as receivers. Because the symmetrization null revealed no actionable directed edge, every controller performed within a few percentage points of the no-action baseline.

Second, for armed-conflict diffusion in the Sahel and Lake Chad regions (six countries, 2012–2023), using the UCDP Georeferenced Event Dataset [11], we predicted that the Malian origin would transmit. Instead, the network proved to be weakly coupled (21% connectedness), the hypothesised origin acted as a net receiver, and all controllers clustered near the baseline. Critically, both failures correspond perfectly to the regularity's null arm, representing either a near-symmetric common shock or a weakly coupled system. Together with the flight-delay case, these outcomes successfully map our framework's empirical scope, demonstrating that directed, exploitable leverage characterizes networks that are directionally asymmetric but vanishes entirely when a system is symmetric or weakly coupled. We note that the conflict analysis represents a confounded null, as it pools two largely decoupled insurgencies at a coarse monthly resolution; we document this specific test in full within the Supplementary Information.

Falsifiability. The regularity maintains strict empirical vulnerability. It would be falsified either by a directionally asymmetric network where transmitter-targeting actively lost, or by a symmetric network where transmitter-targeting won with an advantage that survived the symmetrization null. Neither anomaly occurred. Because directedness is measured uniquely for each case via the symmetrization null, our incorrect predictions reveal errors in our *a priori* classification of a domain's structural bin, not a failure in the fundamental structural link between directedness and controllability. Across our six empirical tests (the held-out COVID-19 network and the five prediction-first trials), the regularity's positive predictions were confirmed three times, refined once, and falsified twice. Out of sample, the framework is demonstrably vulnerable to error, yet it remains robustly accurate for transparent structural reasons wherever a contagion is directionally asymmetric.

When anticipation is needed

A transmitter-targeting controller can be deployed either as a static strategy, where the connectedness table is computed once to guide resource reallocation, or as a dynamic strategy, where the controller continuously replans as the network evolves. The elaborate computational machinery of a learned model and a model-predictive planner is justified only when the simpler static rule fails; we test this boundary condition directly.

On stationary surrogates, the static rule suffices. By decomposing the controller's performance advantage into a static directed-reallocation component and a dynamic-planning component, we demonstrate that the static rule captures essentially all of the theoretical gains on linear surrogates. Specifically, the static rule accounts for 96% of the MPC's total cascade reduction on a synthetic network. On the held-out COVID-19 surrogate, the static rule even slightly outperforms the dynamic planner (+32% versus +28%; Figure 5). Because this evaluation utilizes a decomposition protocol distinct from the benchmark presented in Figure 3, the absolute reduction values are not strictly comparable across the two figures. Rather, our central claim rests upon the consistent, within-experiment ordinal rankings. In short, for first-order, near-stationary systems, the

operational value lies almost entirely in the accurate directed diagnosis rather than the learned optimization machinery.

On non-stationary systems, dynamic anticipation is worth its cost. In sharp contrast, the empirical COVID-19 network is structurally non-stationary in its directed topology. The individual incidence series are stationary in levels (Supplementary Information), so the static full-sample connectedness remains well posed; what is not fixed is the identity of the net-transmitter, which, evaluated over rolling windows, shifts in 72% of iterations, with eight distinct states assuming the leading transmitting role as epidemic waves migrate. In this migratory regime, an adaptive controller that tracks the moving transmitter using strictly historical windows nearly triples the surrogate cascade reduction achieved by the static, full-sample rule (+8% adaptive versus +3% static, compared to a +10% optimization oracle; Figure 6). These findings delineate a robust operational validity map: static directed reallocation suffices for stationary, near-linear systems, whereas dynamic anticipation justifies its computational cost only when the directed topology itself is in motion.

A direct out-of-sample evaluation executed on raw, real-world data strictly independent of the surrogate framework supports this structural map. When the in-sample transmitter is utilized to forecast aggregate downstream stress on unseen testing data, it maintains a positive predictive lead over the loudest unit strictly within systems where the transmitter remains structurally stationary. For instance, it yields an 8% reduction in the held-out aggregate forecast error for the 2008 equity network. Conversely, this predictive lead vanishes entirely in systems where the transmitter migrates, such as epidemic networks or the U.S. housing market, where the net-transmitter shifts dynamically between the bubble and recovery periods.

The predictive lead is distinctly positive in two of the three stationary-transmitter systems, yielding a mean forecast error reduction of 3.3%. However, constrained by the short data samples available in empirical macro-series, this specific result remains statistically non-significant (a three-network sign test yields $p = 0.50$, and a pooled per-unit comparison is similarly non-significant). We therefore report this out-of-sample empirical finding strictly as suggestive corroboration rather than formal confirmation, reserving primary evidentiary weight for the surrogate interdiction results and the pre-committed testing track (Supplementary Information).

Discussion

The robust finding of this study is a falsifiable and transferable structural regularity. Across calibrated surrogates of empirical contagion networks, anticipatory transmitter-targeting outperforms reactive control precisely when, and to the degree that, the contagion is directionally asymmetric. This structural property must be explicitly measured, as it is neither implied by aggregate connectedness magnitude nor reliably anticipated by domain intuition (which failed in two of our explicit tests). Notably, this dissociation between the transmitter and the loudest unit has been independently observed through an entirely separate analytical route in a macro-financial tail-risk network, wherein the real-output sector, rather than the financial hub, emerges as the dominant transmitter [28]. While that study focuses on measurement and forecasting rather than active controllability, it provides convergent corroboration of the core structural insight formalised here.

The pre-committed track record comprising three confirmations, one refinement, and two falsifications constitutes a central empirical contribution of this work. This record elevates the notion that directedness assists control from a plausible heuristic into a formal, falsifiable regularity with a rigorously delineated empirical scope and calibrated out-of-sample support: the directed advantage is

positive across all five prediction-first systems (sign-test $P = 0.031$), and the confirm/refine-versus-falsify partition separates their advantages at an exact permutation $p = 0.10$ ($n = 5$). Indeed, the pre-committed net-transmitter is successfully recovered in five of the seven examined networks, yielding a highly significant Poisson–binomial probability of $P \approx 10^{-4}$ against a uniform random node-selection null. This significance remains robust ($P \approx 1.5 \times 10^{-3}$) even under a stringent restriction to the six networks where the transmitter and the loudest node structurally diverge (Supplementary Information). Ultimately, no individual surrogate control magnitude is offered as a standalone, high-powered confirmatory test. Instead, the evidentiary weight of the regularity rests securely upon three power-independent pillars: the pre-committed ground-truth recovery, the per-case symmetrization and phase-randomisation nulls, and the out-of-sample confirm–refine–falsify testing record. Within this framework, the surrogate effect sizes serve strictly as descriptive empirical illustrations of these underlying structural principles.

Geometric interpretation as conceptual motivation. This work was originally motivated by a geometric interpretation of network dynamics. In this conceptual framing, each unit's context is viewed as a Markov blanket, inter-unit interference represents the leakage of that blanket, and a systemic cascade signifies the total collapse of a separating boundary as the system approaches criticality. This formal apparatus is developed fully in the Supplementary Information, featuring non-Hermitian spectra, optimal transport [18] computed on a learned latent manifold [19], and a discrete-Morse description of the failure surface within the idiom of the free-energy principle and active inference [10, 20]. Because this perspective is demonstrated chiefly on synthetic constructions and a single empirical topological signature, we present it strictly as interpretive scaffolding rather than a validated empirical claim. Crucially, the core empirical results presented in the preceding sections stand entirely independent of this geometric machinery.

Limitations. The reported interdiction magnitudes are properties of a calibrated linear surrogate explicitly rescaled to a supercritical regime ($\rho = 1.05$–1.06). While the qualitative regularity demonstrating that transmitter-targeting outperforms loudest-node control remains strictly invariant to this rescaling, the absolute values and corresponding ratio magnitudes do not. We therefore report these effect sizes strictly as surrogate quantities subject to this standing qualifier, and we emphasise that our framework demonstrates control within a simulated environment rather than through live, real-world interventions.

Several networks are estimated on short temporal samples, such as the flight-delay data where $T \approx 90$ across 18 units. Although ridge regularization and non-negativity constraints ($\Phi_{ij} \geq 0$) actively mitigate overfitting by restricting the effective degrees of freedom, the resulting transmitter rankings can occasionally exhibit fragility in small samples. We report this statistical uncertainty on a per-case basis and verify within the Supplementary Information that the transmitter identities remain highly stable across multi-step forecast horizons ($H \geq 4$). Non-linear, short-series estimators of directed dependence, such as large-scale non-linear Granger causality [27], offer a robust future alternative to our linear diagnostic framework. Furthermore, because variance spillovers can theoretically distort Granger-causal inference [29], the network diagnosis warrants a formal causality-invariance robustness check, particularly concerning the volatility-spillover dynamics observed in the 2008 equity network.

A block-bootstrap analysis confirms that individual controller advantages serve as point estimates bracketed by wide confidence intervals owing to limited data availability. Consequently, only the equity and held-out COVID-19 networks generate intervals that individually exclude zero. The

overarching validity of the regularity therefore rests upon the collective prediction-first pattern, the per-case symmetrization nulls, and the historical ground-truth recoveries, rather than on the isolated magnitude of any single test (Supplementary Information). Additionally, the 1997 currency crisis network is estimated on non-stationary levels, representing a specific methodological limitation that we disclose and quantify in the Supplementary Information. The diagnosed near-unit-root spectral radii primarily reflect elevated system persistence rather than mathematically certified super-to-sub-critical transitions, and they must be interpreted accordingly. Finally, all reported computations are strictly CPU-bound and limited to moderate scales ($N \leq 400$). Expanding this framework to massive-scale networks via GPU implementations remains an active future design target rather than a currently exercised claim.

Outlook. The proposed analytical pipeline is computationally efficient, transparent, and readily transferable across highly diverse domains. The empirical scope delineated by our findings demonstrates that anticipatory interdiction possesses the greatest structural leverage within directed, lower-tier diffusion systems. This applies directly to epidemic, financial, and environmental networks characterized by a quiet, identifiable origin that remains structurally decoupled from its loudest casualty.

Methods

Calibrated surrogates and connectedness. For each empirical system, the multivariate stress series is fitted using a non-negative VAR(1) model:

$$x_t = \Phi x_{t-1} + \varepsilon_t \tag{1}$$

where $\Phi_{ij} \geq 0$ and ridge regularization is applied to stabilise estimation. We assess stationarity on a per-series basis, selecting levels or differences accordingly. Connectedness is quantified via the GFEVD [5, 6], an apparatus belonging to the Granger-causality lineage of directed influence measurement [7]. While this decomposition is typically evaluated at an arbitrarily fixed forecast horizon, we identify the transmitter as the unit exhibiting maximum net (to-minus-from) connectedness. This definition is strictly equivalent to maximizing the horizon-integrated net connectedness:

$$\int \mathrm{net}_i(H)\,\mathrm{d}H \tag{2}$$

This mathematical formulation effectively eliminates the forecast horizon as a free parameter. Across every empirical network recomputed within our pipeline, this integrated metric yields a transmitter identity completely identical to that derived via the fixed-horizon approach (Supplementary Information).

Conversely, the *loudest* unit is defined as the node experiencing the largest realized stress, operationalised per domain as either the cumulative or mean series. Because peak and mean intensities can diverge, this distinction is explicitly tracked; for instance, within the wildfire-smoke network, one state exhibits the highest instantaneous peak while another registers as the loudest by mean accumulation. The spectral radius $\rho(\Phi)$ is continuously monitored across rolling windows. Full model specifications, GFEVD horizon sensitivity analyses, stationarity tests, and bootstrapping procedures are detailed comprehensively in the Supplementary Information.

Throughout this work, the terms "directed" and "transmitter" denote predictive directional influence in this Granger-causal and forecast-error-variance sense, rather than an interventional causal

effect. The interference framing introduced at the outset serves strictly as motivating context rather than a formal potential-outcomes estimand. In this light, the partial- and cross-cluster-interference literature [2, 3, 4] represents a complementary "how large is the effect" paradigm that operates in tandem with our structural "where to act" diagnostic framework.

Interdiction and the symmetrization null. Cascades are simulated upon the surrogate framework, explicitly rescaled to a supercritical regime bounded by $\rho \in \{1.05, 1.06\}$. The evaluated controllers allocate a fixed per-step intervention budget, comparing the no-action baseline, a reactive loudest-node heuristic, correlation centrality, VAR out-strength, impulse-response spillover, static transmitter-targeting, and a dynamic cross-entropy MPC.

To systematically evaluate the role of directedness, the symmetrization null iteratively recomputes the transmitter advantage along a linear interpolation path:

$$\Phi(\alpha) = \alpha\Phi + (1-\alpha)\frac{1}{2}(\Phi + \Phi^\top) \tag{3}$$

evaluated at $\alpha \in \{1, 0.75, 0.5, 0.25, 0\}$. This formulation provides a mathematically rigorous mechanism to vary directedness. Because the matrix $\frac{1}{2}(\Phi + \Phi^\top)$ represents the orthogonal projection of Φ onto the symmetric subspace, the interpolation linearly attenuates exactly the antisymmetric, net-directional component, $\frac{1}{2}(\Phi - \Phi^\top)$, while keeping the symmetric component perfectly fixed. This operation successfully drives every unit's net connectedness to zero at $\alpha = 0$. Crucially, each interpolated matrix $\Phi(\alpha)$ is dynamically re-rescaled to the common spectral radius ρ prior to cascade simulation. This procedure isolates changes in directedness at a fixed level of system criticality, preventing the confounding spectral-radius decay that symmetrization would otherwise induce. Effect sizes are reported as surrogate cascade reductions, and their sensitivity to changes in ρ is detailed in the Supplementary Information.

Prediction-first protocol. For each out-of-domain test, we committed the hypothesised transmitter, the expected loudest-versus-transmitter gap, the unit-selection rule, the estimator, and the explicit falsification criteria to a written, dated record before computing the network. These records are permanently archived and provided alongside our replication code. We label these five initial tests "prediction-first" rather than formally pre-registered because they lack an external, publicly verified timestamp. However, to extend and exercise this protocol prospectively, three additional predictions concerning systems not analyzed in this manuscript have been formally pre-registered with an external, public timestamp prior to analysis (`osf.io/49kn7`).

Statistics and reproducibility. All reported surrogate effect sizes represent mean values computed over 16 fixed random seeds, which are shared symmetrically across controllers and cascade simulations; the per-case symmetrization-null sweeps utilize 10 seeds. Uncertainty regarding the controller advantages is estimated via a stationary block bootstrap, using an average block length of approximately $\sqrt{T}$ across 300 resamples, and is reported via 95% percentile confidence intervals. No data were excluded from the analysis.

Ground-truth origin recovery, operationally defined as an exact match between the pre-committed hypothesis and the empirically identified net-transmitter, is evaluated statistically against both a uniform-random node-selection null and a more stringent restriction that explicitly excludes the loudest node. Detailed data and code accessibility statements are provided below, and all computational analyses adhere strictly to community reproducibility standards for open-science research.

Declarations

Ethics approval and consent to participate. Not applicable. This work uses only publicly available, aggregate data and involves no human participants, human data, or human tissue.

Consent for publication. Not applicable.

Availability of data and materials. All datasets utilized in this study are publicly available and require no access credentials: the Johns Hopkins University COVID-19 case data; the Centers for Disease Control and Prevention (CDC) ILINet influenza-like-illness series via the Delphi Epidata API; the Federal Reserve Economic Data (FRED) exchange-rate series; the Environmental Protection Agency (EPA) AirData daily $PM_{2.5}$ measurements; the U.S. Bureau of Transportation Statistics On-Time Performance database; the Uppsala Conflict Data Program (UCDP) Georeferenced Event Dataset v24.1 [11]; and the national equity-index closing levels (S&P 500, FTSE 100, DAX, CAC 40, Nikkei 225, Hang Seng, AEX, and Bovespa) obtained from their respective index providers.

All analysis code, encompassing data retrieval, surrogate calibration, connectedness calculations, interdiction simulations, the symmetrization null sweeps, every figure, and the prediction-first records, is CPU-only and configured with a pinned software environment and fixed seeds to allow one-command per-figure regeneration. The complete computational pipeline is openly available in a citable, permanent repository at `https://github.com/patrickiben/directed-asymmetry-contagion` and permanently archived under a frozen Zenodo snapshot (`https://doi.org/10.5281/zenodo.21211653`). Exact retrieval scripts and processed series (such as weekly closing levels) are included in the repository, ensuring that every network is fully reproducible without requiring specialised commercial vendor accounts. The dated prediction-first records are deposited within the same archive; because this public deposit post-dates the analyses, its publication date serves as the first external, public timestamp for these records, certifying their existence rather than verifying that each was fixed prior to its respective network computation. The "prediction-first, not pre-registered" status noted in the main text remains the accurate methodological qualifier.

Competing interests. The author declares no competing interests. This is independent research, conducted outside of and unrelated to the author's employment.

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Authors' contributions. P.I. is the sole author and sole contributor: he conceived the study, developed the methods, implemented all software, performed all analyses, and wrote the manuscript. The author read and approved the final manuscript.

Use of artificial intelligence. Generative AI tools assisted the preparation of this work: the literature search, the analysis-code implementation, and manuscript drafting and editing. The author additionally used these tools to help stress-test and error-check the analysis, but all methodological criticism, scientific and editorial decisions, and verification were the author's, who directed each step and is solely responsible for the content. All analyses are fully scripted with fixed random seeds and reproducible from the deposited code; the real datasets were neither generated nor altered by AI, and the dated prediction-first records were authored by the human author. No AI system is an author, consistent with Committee on Publication Ethics (COPE) guidance.

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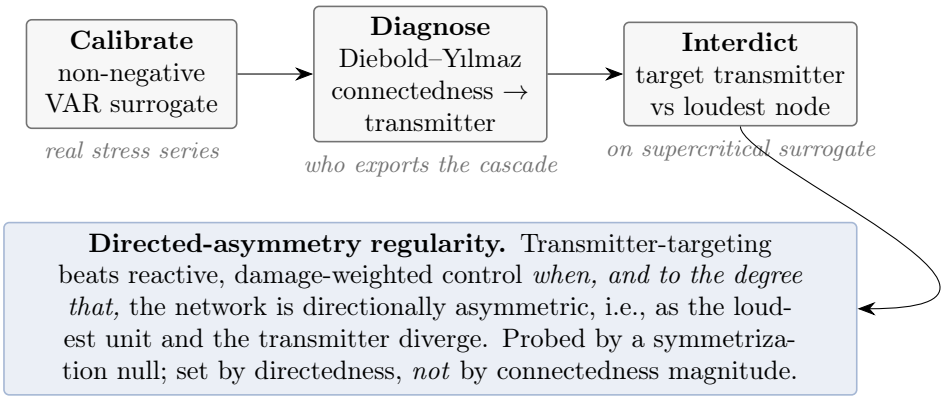


Figure 1: The calibrate–diagnose–interdict pipeline and the regularity. One uniform pipeline is applied to every system: fit a non-negative VAR surrogate to the real stress series, diagnose its directed structure with Diebold–Yılmaz connectedness to identify the transmitter, and interdict on the surrogate (rescaled to a mildly supercritical regime) by targeting the transmitter rather than the loudest-hit unit. The directed-asymmetry regularity is the single regularity that emerges.

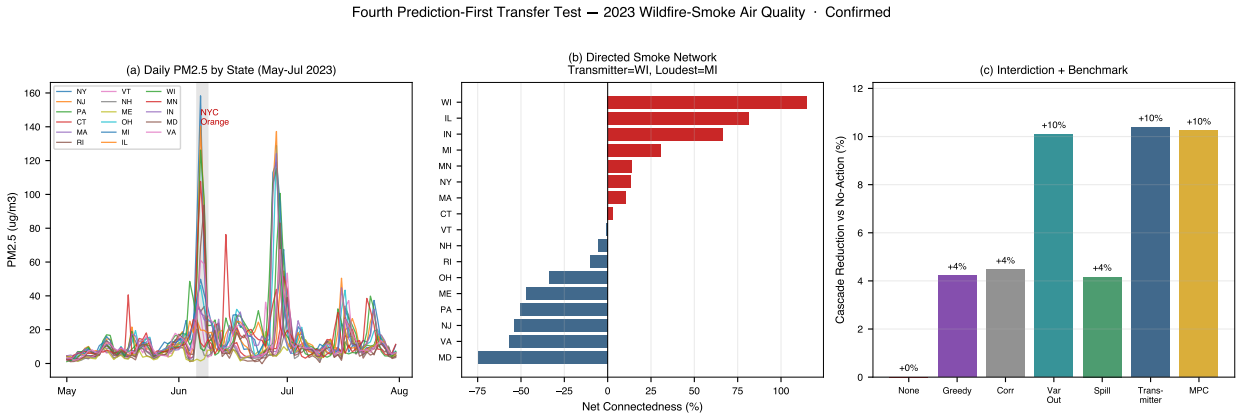


Figure 2: A worked confirmation: the 2023 wildfire-smoke network. **a**, Daily $PM_{2.5}$ across seventeen states. **b**, The directed network: an upwind transmitter (Wisconsin) and a downwind loudest *receiver* (Michigan); the loud-but-receiving structure the regularity requires. **c**, Transmitter-targeting reduces the surrogate cascade $\approx 2.5\times$ more than the loudest-node heuristic; the symmetrization null confirms the advantage is directed. Total connectedness here (82%) is nearly matched by the near-symmetric flight-delay network (77%; Supplementary Information), yet the controller wins here and ties there. Plotted cascade reductions are surrogate point estimates (means over 16 seeds); their block-bootstrap 95% intervals are wide and, for every network here except the 2008 equity crash and the held-out COVID-19 network, include zero (Supplementary Information). The directed advantage is established by the symmetrization null, not by the magnitude.

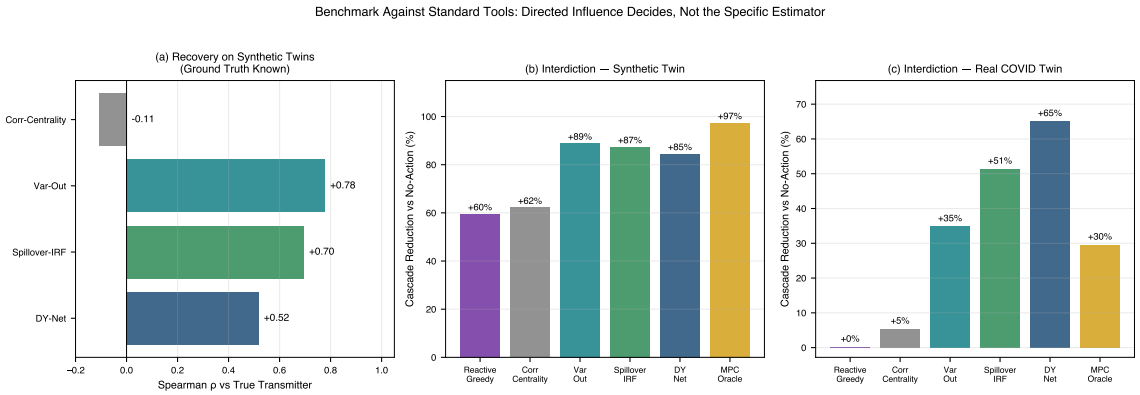


Figure 3: Benchmark against standard estimators on a held-out COVID-19 network. Directed influence measures (Diebold–Yılmaz net connectedness, impulse-response spillover, VAR out-strength) reduce the surrogate cascade by 35–65% and recover the true support targets (rank correlations 0.5–0.8), against 5%/−0.1 for undirected correlation centrality and 0.2% for reactive triage. Directedness, not a particular estimator, is responsible for the result.

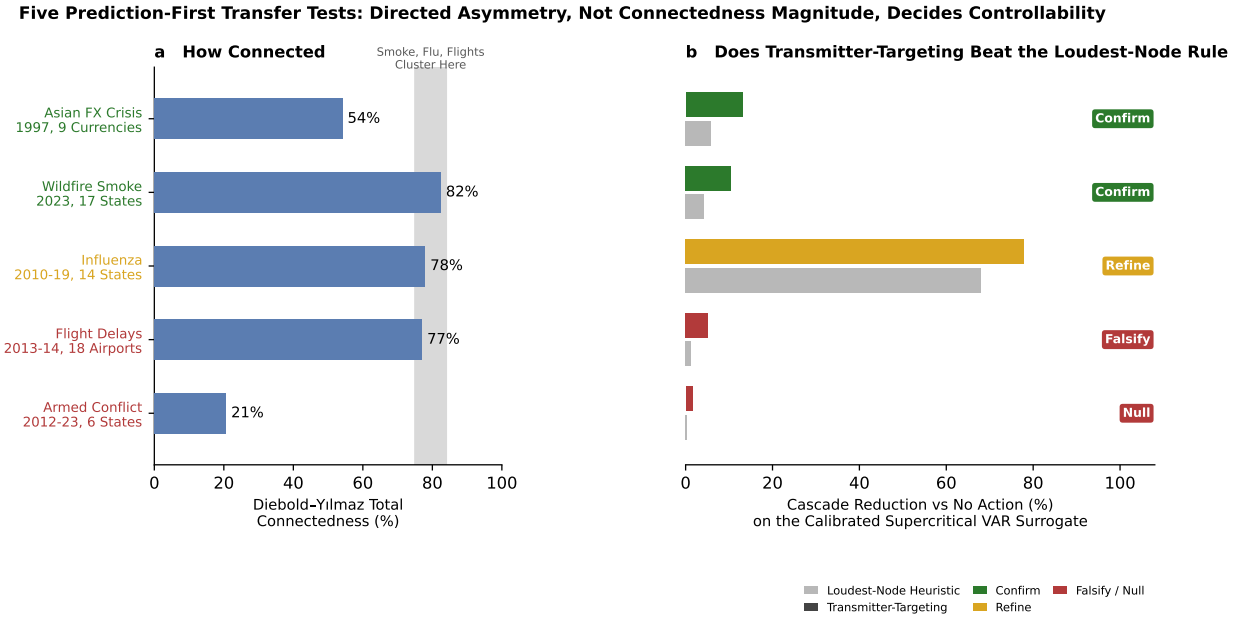


Figure 4: Five prediction-first out-of-domain transfer tests. **a**, Diebold–Yılmaz total connectedness; smoke (82%), influenza (78%) and flight delays (77%) cluster (gray band) yet diverge in outcome — connectedness magnitude does not predict controllability. **b**, Surrogate cascade reduction for the reactive loudest-node heuristic (gray) versus transmitter-targeting (colored), with the committed verdict. The two confirmations (Asian FX crisis, wildfire smoke) show a clear transmitter advantage; influenza *refines* the regularity (transmitter = loudest, so the gap and the advantage are small); flight delays and armed conflict *falsify* committed predictions (near-symmetric and weakly-coupled respectively, with every controller near the no-action baseline). All reductions are on the calibrated supercritical VAR surrogate, and are surrogate point estimates whose block-bootstrap 95% intervals are wide and include zero for every twin shown (only the equity and held-out COVID-19 networks individually exclude zero; Supplementary Information). The regularity rests on the symmetrization null, ground-truth recovery, and prediction-first record, not on these per-twin magnitudes.

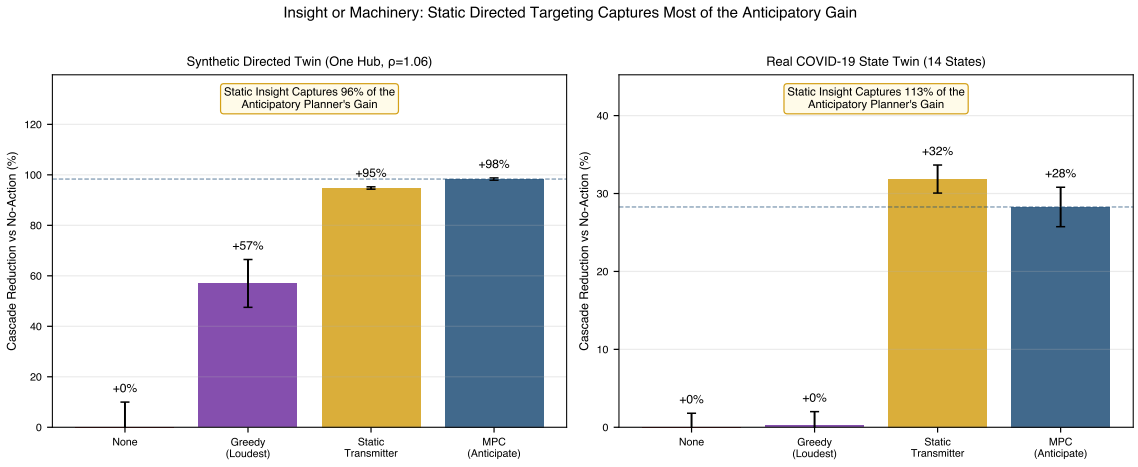


Figure 5: Insight versus machinery. The controller's advantage decomposed into a static directed-reallocation component and a dynamic model-predictive component. On a synthetic network, the static rule captures 96% of the planner's reduction; on the held-out COVID-19 surrogate it slightly exceeds it (+32% vs +28%). On near-stationary, first-order systems the value is the directed diagnosis, not the learned planner.

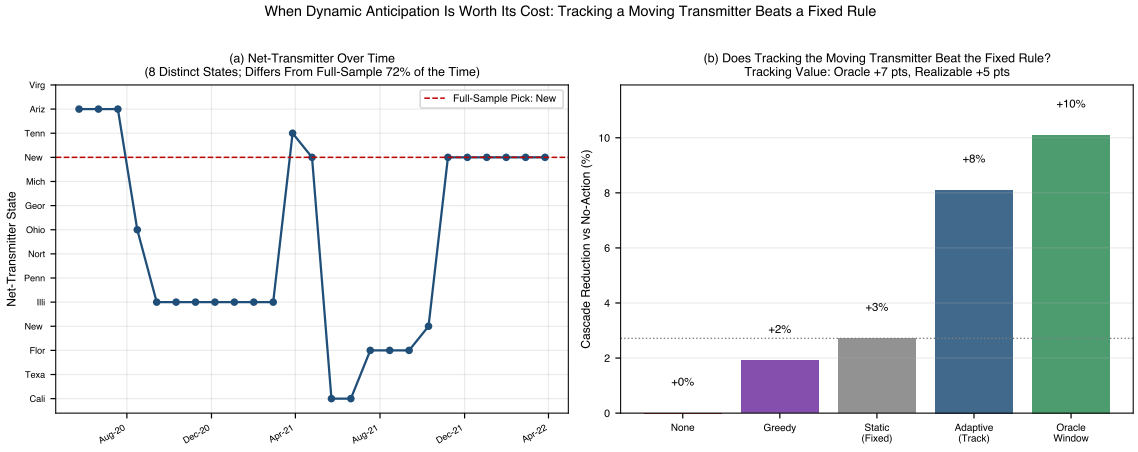


Figure 6: When dynamic anticipation is worth its cost. On the real COVID-19 network, the net-transmitter is non-stationary (it shifts in 72% of rolling windows; eight states take the lead across waves). A controller that tracks the moving transmitter using only past windows nearly triples the static full-sample rule (+8% adaptive vs +3% static, +10% oracle). Dynamic anticipation is justified only where the directed structure itself moves.